MILITARY SPECIFICATION

BATTERY, RECHARGEABLE, NICKEL-CADMIUM, VENTED, AIRCRAFT, BB-716/A

1. SCOPE

1.1 Scope. This specification covers the requirements for BB-716/A vented, rechargeable nickel cadmium aircraft batteries and cells and harnesses used therein.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6140

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SPECIFICATIONS

FEDERAL

FF-N-836	- Nut, Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat.
FF-S-92	- Screw, Machine, Slotted, Cross Recessed or Hexagonal Head.
QQ-C-502	- Copper Rods and Shapes; and Flat Products with Finished Edges (Flat Wire, Strips and Bars).
QQ-C-576	- Copper Flat Products with Slit, Slit and Edge Rolled, Sheared, Sawed or Machined Edges (Plate, Bar, Sheet and Strip).
QQ-N-290	- Nickel Plating (Electrodeposited).

MILITARY

MIL-M-14	- Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-W-4088	- Webbing, Woven, Textile, Nylon.
MIL-M-13231(ER)	- Marking of Electronic Items.
MIL-C-16173	- Corrosion Preventive Compound, Solvent Cutback, Cold Application.
MIL-W-16878	- Wire, Electrical Insulated.
MIL-I-16923	- Insulation Compound, Electrical Embedding.
MIL-S-19500/260	- Semiconductor Device, Diode, Silicon, Power Rectifier Types 1N1202A, 1N1204A, 1N1206A, 1N3671A and 1N3673A (Both A and RA) JAN, JANTXV, and JANS.
MIL-T-23648	- Thermistor (Thermally Sensitive Resistor), Insulated.
MIL-S-24236/1	- Switches, Thermostatic (Bimetallic), Subminiature, Type I, Hermetically Sealed, Single Pole, Single Throw.
MIL-P-47104(MI)	- Potting Compound, Epoxy Resin Thermosetting.
MIL-B-81757	- Batteries and Cells, Storage Nickel-Cadmium, Aircraft, General Specification for.

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STANDARDS

FEDERAL

FED-STD-595	- Colors.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-143	- Standards and Specifications, Order of Precedence for the Selection of.
MIL-STD-202	- Test Methods for Electronic and Electric Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MS3474	- Connectors, Receptacle, Electric Series 2, Crimp Type, Rear Mounting, Jam Nut, Bayonet Coupling, Classes A. L. S and W.
MS3509	- Receptacle, Electric, Aircraft Storage Battery.
MIL-STD-45662	- Calibration System Requirements.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.2 Other Government documents, drawings and publications. The following Government documents, drawings, and publications form a part of this specification to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D4066-82 Nylon, Injection and Extrusion Materials (PA)

(Applications for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation.

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification No. A-3 and No. 14

(Application for copies should be addressed to the American Trucking Association, 1424 26th Street N.W., Washington, DC 20009)

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules (Application for copies of these freight classification rules should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, Illinois).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and applicable specification sheet, the latter shall govern.

3.1.1 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.2 First article requirements. The battery, cells and harness furnished under this specification shall be a product which has been tested and has passed the First Article inspection and testing specified herein. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.8).

3.3 Materials and components. When a definite material or component is specified, it shall be in accordance with the requirement listed in Table II. When deemed necessary by the government, certification from the source of the material or component will be required. In the absence of certification from

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the source, a certificate of analysis or certified inspection data shall be required (see 4.3.1).

3.3.1 Metals. External metal or component parts of each battery, cell and harness shall show no evidence of cracking, chipping, scaling, corrosion or other deleterious effects when exposed to 1.300±0.025 specific gravity KOH electrolyte solution.

3.3.2 Plastic materials. Plastic materials used in each battery, cell and harness shall show no evidences of cracks, blisters or other deterioration and shall not change more than 2.5 percent in weight or 2.0 percent in any dimension when exposed to 1.300±0.025 specific gravity KOH electrolyte solution.

3.3.3 Elastomer materials. All elastomer materials used in each battery, cell and harness shall show no evidence of cracks, blisters or other deterioration after being exposed to 1.300± 0.025 specific gravity KOH electrolyte solution. Materials of pads, gaskets and O-rings shall be selected to provide adequate seal and minimize compression set.

3.4 Design and construction. The battery, cells and harness shall conform to this specification and the applicable specification sheets.

3.4.1 Batteries.

3.4.1.1 Battery container and cover. The battery container and cover shall be of alkali-resistant high strength steel, free of rough spots, pits, blow holes and other deformations. The container and/or cover shall be completely coated by an epoxy resin to a minimum thickness of .15±.008 inches. The dimensions and location of receptacles, hold-downs, latches, latch hooks, vent tubes, name and instruction plates, and carrying strap shall conform to the applicable specification sheets. The cover shall be fitted with a hold-down pad having an integral elastomer gasket positioned so as to match the rim of the container body for a gas-tight fit between the cover and the container body. The seal shall prevent the loss of pressure in excess of 2.75 inches of water when tested in accordance with 4.7.8.

3.4.1.2 Latches. The cover shall be secured to the container body by latches mounted on the container body and latch hooks mounted on the cover. The latches shall conform to the Corbin Cabinet Lock Division, American Hardware, Item No. 15834, or Nielson Latch No. 5C-B-83314-2SS or equal and the latches and hooks shall be mounted on the cover and container body as specified in the applicable specification sheets (see 3.1).

3.4.1.3 Vents. The container body shall be provided with tubular vents to permit gases liberated by the cells of the

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battery to be purged. These vents shall be located as shown in the applicable specification sheets (see 3.1).

3.4.1.4 Main power receptacle. The battery shall utilize a two pin receptacle for a main power connector. The receptacle shall be mounted as shown in the applicable specification sheets and shall conform to MS3509. The receptacle shall be capable of withstanding a pull out force of 250±25 pounds and a torsional force of 6.5 foot-pounds.

3.4.1.5 Container liner. Container liners, when used, shall be 0.015 inch or multiples of same, polyamide (nylon) per ASTM D4066-82, Class PA111. Liners shall be cut to allow rapid battery assembly and disassembly, but deter cells from moving in relation to the battery container when the battery is vibrated.

3.4.1.6 Intercell Connectors. Intercell connectors shall comply with the applicable specification sheets.

3.4.2 Cells. The battery shall contain the number and type of cells in the configurations specified by the applicable specification sheets.

3.4.2.1 Cell mounting. The cells shall be removable but securely mounted in the battery container such that they are not adversely affected by tests performed on the battery to determine its compliance with the requirements of this specification.

3.4.2.2 Cell container. The cell container shall be made of high strength, non-porous, alkali-resistant polyamide. The surfaces of containers shall have a smooth finish, free from pits, blow-holes, rough spots, or other deformations.

3.4.2.3 Cell terminals. The size and spacing shall be as specified in the applicable specification sheets. The contact surfaces of the terminal shall not be obstructed by insulating compounds.

3.4.2.4 Internal pressure. Each cell shall be assembled and constructed in such a manner that it shall be capable of withstanding an internal pressure of 20±1 psig for 15 seconds.

3.4.2.5 Cell vents. Each cell shall be equipped with a vent per applicable specification sheet. The vent plug, when installed to the cell at a torque of 3.5±1.0 in-lb, shall form an airtight seal for the cell when the cell internal pressure is below 2 psi. The vent plug shall be capable of providing automatic relief when the internal cell pressure reaches 6±4 psi. The vent plug shall withstand 10 in-lb of torque with no damage.

3.4.2.6 Potassium carbonate. Cells from the battery shall be tested for carbonate content in the electrolyte. The electrolyte

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shall not contain more than 60 grams per liter of carbonate.

3.4.3 Harness assembly. The harness assembly shall conform to the applicable specification sheet.

3.4.3.1 Thermistors. The harness assembly shall utilize two temperature sensor devices with a resistance output conforming to the applicable specification sheet. If a thermistor is used it shall conform to MIL-T-23648. The thermistor or device shall be wired per applicable specification sheet. Potting employed to isolate the thermal sensors from any battery voltage shall conform to 3.6.6.

3.4.3.2 Sensor receptacle. The battery shall use a MS3474-L12-10PN receptacle to provide connection to battery sensing and control points. The location of the receptacle shall conform to applicable specification sheets (see 3.1).

3.4.3.3 Thermoswitches. The harness assembly shall have two temperature cutoff devices (thermoswitches) connected in series to provide a signal for high temperature conditions. The thermoswitches shall conform to MIL-S-24236/1-SKKKG. The switches shall open when the temperature is $160\pm5^{\circ}F$ ($71\pm2.8^{\circ}C$) and shall close when the temperature drops to $145\pm5^{\circ}F$ ($62.8\pm2.8^{\circ}C$). The thermoswitch contacts shall be isolated from any battery voltage. Potting used to encapsulate the thermoswitches shall conform to 3.6.6.

3.4.3.4 Device mounting. The devices (thermoswitches and thermistors) shall be mounted on a custom or standard connector strap in such a manner that it shall not become detached during handling. The devices shall be positioned on the connector in such a manner that it shall not interfere with the tool operations required for removal and mounting of cell. If the mounting scheme does not require a connector strap, the manufacturer shall provide a separate M49450/1-1 connector strap for each device that is not mounted on a connector.

3.4.3.5 Insulation. All current carrying parts of the harness assembly shall be insulated from the metal parts of the sensor receptacle except for the applicable pins. The insulation resistance between any current carrying part and the sensor receptacle (excluding the applicable pins) shall be above one megohm.

3.4.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be employed in intimate contact with each other. Dissimilar metals are defined by MIL-STD-889.

3.4.5 Corrosion prevention. After the battery has been assembled, all exposed metal surfaces of the cells, intercell connectors and associated hardware, and harness shall be coated

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with a corrosion preventive compound which meets the requirements of MIL-C-16173, Grade 4. Care shall be taken not to expose rubber vent sleeves, vent openings or current carrying surfaces to the corrosion preventive film. Coating shall be applied evenly and without voids.

3.4.6 Cleaning. All parts of the battery, cells and harness when assembled shall be cleaned in accordance with good commercial practice. Corrosive material shall be removed completely. After assembly the battery, cells and harness shall be cleaned thoroughly and shall be free from particles of solder, flux and other foreign material.

3.4.7 Workmanship. All batteries, cells and harness shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect their life, serviceability, or appearance in accordance with Requirement 9 of MIL-STD-454.

3.4.8 Interchangeability. Parts and components of the batteries, cells and harness shall be functionally and dimensionally interchangeable.

3.4.9 Shipping condition of cells. All cells whether shipped individually or within the batteries shall be shipped complete with electrolyte to proper level and completely discharged, but fully formed. During shipment, a shorting plug or spring shall be connected to the battery's main power receptacle or to the cell's terminals (if shipped separately).

3.4.9.1 Cell electrolyte level. Each cell shall have the electrolyte level adjusted so that after a full charge and 2 hours stand, the level shall be 1/8 to 1/4 inches above the baffle.

3.4.9.2 Low capacity cell electrolyte level. Each low capacity cell shall have the electrolyte level adjusted after a full charge and 2 hour stand; the level shall be 1/8 to 1/4 inches above the baffle (equivalent to 1.06 to 1.16 inches above the plates).

3.4.10 Carrying Strap. The battery cover shall have a carrying strap made of web nylon per MIL-W-4088. The strap shall be riveted on to the battery per Requirement 12, MIL-STD-454 or a strap connector shall be welded on to the battery per Requirement 13, MIL-STD-454.

3.4.11 Diode. The battery shall have a 1N1202RA diode connected across the low capacity cell as a bypass to prevent overdischarge. The diode shall conform to the requirements of MIL-S-19500/260.

3.5 Color and marking.

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3.5.1 Color. The color of the container and cover except latches, hold-down hooks, other external hardware, pads, and identification markings of the battery shall conform to FED-STD-595 within the range of blue color chips 15090 to 15193.

3.5.2 Marking. All markings and identifications shall be permanent and shall be resistant to 1.300±.025 specific gravity KOH electrolyte. The markings and identification shall conform to MIL-M-13231. The location of each marking shall conform to the applicable specification sheet (see 3.1).

3.5.2.1 Cell terminal marking. Positive terminal markings shall be impressed or embossed with a (+). Negative terminal markings shall be impressed or embossed with a (-). The polarity marking shall be as close as possible to the terminals on the cover of the cell. The positive half of the cell cover shall be permanently marked red. Maximum width of the red band shall be no less than 0.5 inches.

3.5.2.2 Cell identification. Each cell shall have the following information printed in black, 1/8 inch lettering:

Battery, Rechargeable * (Cell)
*(NSN).
Part of BB-716/A
*(AH)/ 1HR/ 80 F/ 0.95 V
*(CONTRACT NO.)
*(MANUFACTURER PART NO.)

* FILL IN THE APPLICABLE INFORMATION

Example label:

Battery, Rechargeable (Cell)
NSN 6140-01-105-6511
Part of BB-716/A
3.6 AH/ 1HR/ 80 F/ 0.95V
(Contract No.)
(Manufacturer part no.)

Information shall be marked on one of the two largest vertical surfaces of the cell.

3.5.2.3 Battery identification. Each battery shall have its identification marking silk screened with an epoxy overlay or equal. The markings shall be printed with white, 1/8 inch lettering and shall have the following information:

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Battery, Rechargeable
BB-716/A
5.5 AH/ 1HR/ 80 F/ 18 volts
*(Contract No.)
*(Date code)
*(Serial No.)
NSN 6140-01-089-8134
*(Name of manufacturer)
Maximum weight 17.5 pounds

* Fill in applicable information

3.5.2.4 Caution label. This label shall have black letters on a yellow background per MIL-M-13231. The label shall contain the following lettering size and information:

CAUTION

DO NOT USE ACID .200±.030 inch lettering

ELECTROLYTE

Acid will permanently damage this battery.
Avoid using tools which have been used on .120±.030 inch lettering
acid type storage batteries. Consult
technical manual before servicing.

3.5.2.5 Date code. The date code shall indicate the month and year of manufacture of the battery and cells by means of a four digit number in which the first two digits indicate the month of the year and the last two digits indicate the year. Months earlier than the tenth shall be indicated by a single digit preceded by "0". When the manufacturer of a battery and cell is completed during the last three working days of the month, or the first three working days of the subsequent month, the manufacturer is permitted to use either month as the coded month of manufacture. The date code shall be located on the cover of the cell. The numbers shall be 1/8 inches in height and black in color. For the low capacity cell the numbers shall be white in color.

3.5.2.6 Battery polarity marking. The polarity shall be conspicuously and durably marked on the battery container and adjacent to the terminals.

3.5.2.7 Battery warning marking. This marking shall be in white print and read as follows:

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WARNING

1. Area around battery must be well ventilated and cover removed during charge.

2. Do not allow caustic electrolyte to contact body, clothing, tools, etc. If contact is made wash immediately with water. Neutralize with 3% boric acid or vinegar. Rinse with water but do not allow these fluids to enter cells. CAUTION FLUSH EYES WITH WATER ONLY.

3.5.2.8 Instruction marking. This marking shall be in white and read as follows:

INSTRUCTION

1. Remove cover and brush off white carbonate deposits with vents in place.

2. Remove shorting plug or spring.

3. Refer to charging instructions in applicable technical manual:

4. After charge, allow 2 hour stand, remove cell vents and adjust electrolyte using distilled water to 1/4 inches above plate for the cells and 1 inch above plate for the low capacity cell. Replace cell vents and cover.

5. Periodically soak cell vents in distilled water.

3.5.2.9 Warning Tags. Where shipping devices are furnished by the contractor to assure proper shorting, disallow cell leakage etc., alkali resistant tags with strong alkali resistant, non-metallic cord shall indicate action or actions to be taken by field personnel to remove and/or replace same.

3.5.2.10 Connector marking. This marking shall be in white and centered over the terminal location. The arrows shall point vertically down on each terminal pin and read as follows:

CONNECTOR

3.6 Performance.

3.6.1 Capacity. When specified, the battery and cells shall be discharged at the current rate specified in the applicable specification sheet to the specified cut-off voltage for the test listed. The battery and cell capacity shall be equal to the specified discharge current (amperes) multiplied by the discharge time above the specified cut-off voltage (hours). The battery and cells shall provide the capacity specified in the applicable specification sheets.

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3.6.2 Environmental testing. The battery and cells when subjected to the tests of Table I shall show none of the following:

a. Dimensional distortion beyond specified limits or cracking of case or cover of either cell or battery.

b. Radical current or voltage fluctuations during any test.

c. Mechanical failure of any part.

d. Electrolyte leakage or any spilling of electrolyte at any time during test. Venting of gasses containing entrained electrolyte is permitted during charge.

e. Breakdown of insulation, stripping of metal plating from any component part, corrosion of metal parts, or loosening of protective coating from the battery container or cover.

f. Deterioration of battery/cell identification markings.

3.6.2.1 Harness testing. The harness assembly when subjected to the temperature shock test of 4.7.23 or alkali resistance test of 4.7.21 shall show none of the following:

a. Breakdown of insulation, stripping of metal plating from connectors, corrosion of metal parts or cracking of potting compound.

b. Thermistors fail to meet the requirements of 3.4.3.1.

c. Thermoswitches fail to meet the requirements of 3.4.3.3.

d. Mechanical failure of components.

e. Dimensional distortion beyond specified limits.

3.6.3 Electrolyte leakage. The battery and cells shall show no evidence of electrolyte leakage during the performance of any of the tests specified in Section 4. Venting of gasses containing entrained electrolyte is permitted during charge.

3.6.4 Operating position. The battery shall be capable of operating in an inverted position without loss of any electrolyte from the cells. The cells shall be capable of operating in an inverted position without loss of electrolyte.

3.6.5 Insulation. All current carrying parts shall be insulated from the battery container. The insulation resistance between any current carrying part and the battery container shall be above one megohm.

3.6.6 Potting. Compounds used for potting shall not support

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flame, absorb moisture, or be damaged by battery electrolyte. The potting compound shall conform to MIL-P-47104(MI) or MIL-I-16923(STYCAST 2651 or equal).

3.6.7 Wiring and electrical connections. All wires used shall be teflon insulated and shall conform to MIL-W-16878. The wire size, length and insulation colors shall be in accordance with the applicable specification sheet. All electrical connections shall be potted per 3.6.6 and shall conform to the applicable specification sheet (see 3.1).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Inspection covered by subsidiary documents including components and materials inspection (see 4.3 and 4.3.1).

b. First article inspection (see 4.4).

c. Quality conformance inspection (see 4.5).

1. Quality conformance inspection of product before packagin (see 4.5.1).

2. Packaging inspection (see 4.8).

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4.3 Inspection covered by subsidiary documents. Inspections required under the applicable subsidiary documents form part of the inspections required by this specification. (See Table II).

4.4 First article inspection. This inspection shall be performed by the contractor unless otherwise specified in the contract. The inspection shall consist of the following:

a. Inspection and testing of batteries specified in Table IV and shall be performed in the order indicated.

b. Inspection and testing of three samples of each applicable battery components and materials treated and processed as they would be for fabrication of finished battery in accordance with Table III. All samples shall be marked properly with identifying information.

4.4.1 First article compliance report. The contractor shall submit a compliance report with any battery, harness, or cell type submitted for acceptance. The report, complete in detail, shall be submitted regardless of whether the tests to determine compliance with specification requirements are conducted at the contractor's facilities, government facilities or independent facilities. The contractor's report shall indicate to what degree the equipment complies with each individual electrical, mechanical, or other specification requirements. The degree of compliance shall be expressed in the report by the use of the same terms as those in the specification to express the requirement (see 6.6).

4.4.2 First article samples. Unless specified, the contractor shall furnish four first article samples selected at random of each battery type on order.

4.4.2.1 Separate cell procurements. For separate cell procurements, the contractor shall furnish 76 first article samples selected at random. The cells shall be organized into four equal number cell groups (19 cells each) for inspection purposes.

4.4.2.2 Separate low capacity cell procurements. For separate low capacity cell procurements, the contractor shall furnish four first article samples selected at random.

4.4.2.3 Battery plus component procurements. For procurements of batteries plus components (cells, low capacity cells and/or harnesses) from the same manufacturer, the batteries shall be subjected to first article sampling per 4.4.2. and the harness shall be subjected to first article sampling per 4.4.2.4. The batteries shall qualify cells and low capacity cells. The harness shall also be qualified.

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4.4.2.4 Separate harness procurements. For separate harness procurements, the contractor shall furnish three first article samples selected at random.

4.4.3 Definition of failure. First article sample failure is defined as follows:

a. Failure to deliver the electrical performance specified in 3.6.1.

b. Distortion, cracking or breakage which prevents the battery or any of its components from mating with a corresponding part or maintaining its electrical leak proof requirements.

4.4.3.1 Degree of failure. The batteries, harnesses, or cells shall be considered unacceptable if any of its first article samples fail to pass the inspections of Table III or IV.

4.4.3.2 Rework and retest provisions. Once a failure has occurred, no rerun can be made of that test on the same or a new sample from the same group of samples unless positive action consisting of design change, modification, or rework, actually related to the observed failures has been applied to the group. Any subsequent rerun must start at the beginning of the test in which the failure occurred provided aforementioned action will not affect any tests prior to the test in which the failure occurred.

4.5 Quality conformance inspection.

4.5.1 Quality conformance inspection before packaging. Inspection of product delivery shall consist of groups A, B, and C inspections. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government shall review and evaluate the contractor's inspection records. In addition, the government at its discretion may perform all or part of the specified inspection, to verify the contractor's compliance with specified requirements (see 6.5). Test equipment for government verification inspection shall be made available by the contractor.

4.5.2 Inspection lot definition. An inspection lot shall be defined as the quantity of batteries, cells and harnesses of any one type produced at any one place of manufacture on any one contract, submitted at one time to quality conformance inspection. Batteries or cells with no more than two consecutive date codes shall appear in a particular inspection lot. A specific date code shall not be found in any more than one inspection lot.

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4.5.2.1 Separate cell procurements. For separate cell procurements, a cell inspection lot shall be defined as the number of cells produced divided by 19 to determine lot size for inspections. Each group of 19 cells shall be considered as a battery for inspection purposes.

4.5.2.2 Separate low capacity cell procurements. For separate low capacity cell procurements, a low capacity cell shall be considered as a battery for inspection purposes.

4.5.2.3 Separate harness procurements. For separate harness procurements, a harness shall be considered as a battery for inspection purposes.

4.5.2.4 Battery plus component procurements. For procurements of batteries plus component parts (cells, low capacity cells and harness assemblies) from the same manufacturer, only the batteries shall be subjected to Group A, B and C inspections. The batteries shall qualify all.

4.5.3 Group A inspection. Each unit of each inspection lot shall be subjected to and meet the requirements of Table V and VI. Defective units shall be individually rejected. Group A inspection shall be performed in any order which is satisfactory to the government.

4.5.3.1 Disposition of Group A samples. Unless otherwise specified, Group A samples shall be shipped and included in the quantity on order if they meet the necessary requirements.

4.5.4 Group B inspection. This inspection, including sampling, shall conform to Table VII and procedures of MIL-STD-105 with tests of each subgroup conducted in the order shown on the same samples. Group B inspection shall be performed on inspection lots that have passed Group A inspection. Samples for subgroup II are to be selected from units that have passed subgroup I inspection, and the remainder of subgroup I samples may be furnished under contract. No shipment shall be made until Group B inspection is completed and approved by the government inspector.

4.5.4.1 Disposition of Group B samples. Unless otherwise specified, sample units which have been subjected to and passed Group B inspection may be shipped against the quantity on order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.5.5 Group C inspection. Group C inspection shall be performed in accordance with Table VIII on two (2) units that have been randomly selected from the lot. One unit shall be assigned to subgroup 1 one to subgroup 2. The samples may be selected before

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full marking and group A inspection has been applied to the lot. However, the test items shall be fully completed prior to the start of Group C testing. Shipment of batteries or cells shall not be made until subgroups I have met the required inspections and 100 cycles of the required number of cycles of subgroups II have been successfully completed. Inspection shall be performed in the order shown.

4.5.5.1 Disposition of Group C samples. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection shall not be delivered on the contract or purchase order.

4.5.5.2 Noncompliance. If a sample fails to pass Group C inspection, the manufacturer shall notify the qualifying activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After corrective action has been taken Group C inspection shall be repeated on additional sample units (all tests and examinations, or the tests which the original sample failed, at the option of the qualifying activity). Groups A and B inspection may be reinstituted; however, final acceptance and shipment shall be withheld until the Group C inspection has shown that the corrective action was successful. In the event of failure reinspection information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.6 Inspection conditions and equipment.

4.6.1 Inspection and test conditions.

4.6.1.1 Normal temperature. Unless otherwise specified, all measurements and tests shall be made at 73±5°F (23±5°C) at ambient atmospheric pressure and relative humidity with a minimum storage of 30 minutes between charge and discharge.

4.6.1.2 Charging.

4.6.1.2.1 Constant current method. The batteries shall be charged constant current at the 5.5 amperes to 1.55±.01 volts per cell then 1.1 amperes for another 3 hours. Unless specified otherwise, the electrolyte shall be adjusted with distilded water to the levels specified by 3.4.9.1 and 3.4.9.2 two hours after charge termination.

4.6.1.2.2 Fast charge method. The batteries shall be charged at 7.0 amperes to 1.55±.01 volts per cell, then continue for

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another minute at the same rate.

4.6.1.3 Reconditioning. If a sample fails to meet the specified capacity or voltage requirement, it shall be subjected to reconditioning prior to repeating the capacity or voltage test (rerun). Reconditioning shall consist of deep discharge on an individual cell basis for batteries or individual cells. Each cell shall be discharged to zero volts within 24 or less hours, followed by a full charge (see 4.6.1.2) and adjustment of electrolyte level. Should the sample not meet the capacity or voltage requirement upon rerun, the repeated test shall be considered a failure. If the sample passes rerun, the specified test schedule shall be resumed.

4.6.1.4 Low capacity cell. For all discharges conducted on the battery in the tests specified in this specification and applicable specification sheets, the voltage and length of discharge prior to the cut-off voltage specified of the low capacity cell shall be monitored. The low capacity cell for all discharges shall meet the requirements of 3.6.1.

4.6.2 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of required inspection. Test equipment for government verification inspection shall be made available by the supplier. The contractor shall establish and maintain a system for the calibration of all measuring and test equipment in accordance with MIL-STD-45662.

4.6.3 Instrument Accuracy.

4.6.3.1 Electrical indicating instruments. All voltmeters and ammeters shall be accurate within 0.5 percent of the full scale reading. The range shallbe such that the readings are taken on the upper half of the scale. Timers shall be accurate within 0.5 percent. The sensitivity of voltmeter shall be at least 1000 ohms per volt.

4.6.3.2 Resistance tolerances. In all tests involving discharge through a resistance, the total circuit resistance shall be accurate within 0.5 percent.

4.7 Examination and test methods.

4.7.1 Corrosion resistance. Samples of all metal parts of the battery shall comply with 3.3.1 after being subjected to the Corrosion Resistance Test (para 4.6.1) of MIL-B-81757.

4.7.2 Alkali absorption of plastic material. Samples of plastic materials shall comply with 3.3.2 after being subjected to the Alkali Absorption of Plastic Material Test (para 4.6.2) of MIL-B-81757. The identification markings on the plastic materials

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shall comply with 3.5.2.

4.7.3 Alkali resistance of elastomer material. Three samples each of pads, gaskets, O-rings, etc., made of elastomer material used in the battery shall be subjected to the Alkali Resistance of Elastomer Material Test (para 4.6.3) of MIL-B-81757. The elastomer shall comply with 3.3.3.

4.7.4 Vent plug test.

4.7.4.1 Venting tests. The vent of each cell being tested shall be installed in an adapter through which controllable air pressure may be applied to the vent. The air pressure shall be raised from 0 to 10 psig and returned to 0 psig in a steady and uniform manner over a 2.0 minute period. The test shall be performed at ambient air temperature and repeated at -30°C (-22°F) except that the vent shall be conditioned at -30°C (-22°F) in air for 48 hours prior to testing. The cell vents shall conform to 3.4.2.5.

4.7.4.2 Torque test. The vent plug of each cell beingtested shall be closed to a torque of 10 in-lb and then opened. This closing and opening shall be repeated a total of five times and the vent plug shall meet the requirements of 3.4.2.5.

4.7.5 Visual and mechanical. Samples shall be examined to verify that the basic materials, component materials and parts, design and construction, marking and workmanship are in accordance with all requirements of 3.4, 3.5, 3.6.3 and Table VI.

4.7.6 Insulation resistance. Insulation resistance of a discharged battery shall be measured between the battery terminal and the exposed metal of the latch, and between sensor receptacle contact pins E or F and the exposed metal of the latch. The insulation resistance measurement shall be made in accordance with MIL-STD-202, Method 302, Test Condition B. Each measurement shall last 20 seconds and shall meet the requirements of 3.6.5.

4.7.7 Dimension and weight. The battery shall be examined to verify compliance with the requirements of 3.4.

4.7.8 Container body to container cover seal. The battery shall be subjected to the Container Body to Container Cover Seal Test (para 4.6.8) of MIL-B-81757. The battery shall meet requirements of 3.4.1.1.

4.7.9 Potassium carbonate. The sample cell shall be subjected to the Potassium Carbonate Test (para 4.6.10) of MIL-B-81757. The cell shall comply with 3.4.2.6.

4.7.10 Capacity. The battery shall be charged per 4.6.1.2.1 and discharged at the rate specified by the applicable specification

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sheet to the cut-off voltage specified. The battery shall meet the requirements of 3.6.1 and 3.6.3.

4.7.10.1 Initial capacity. The battery shall be charged per 4.6.1.2.1 and shall reach an average voltage level of 1.55±.01 volts/cell within 1 hour and 40 minutes of the start of charging in order to demonstrate conformance with the requirement for full formation of the cells (para 3.4.9). After 2 hours rest, the electrolyte level of the cells and low capacity cell shall meet the requirements of 3.4.9.1 and 3.4.9.2. The battery shall be discharged at the rate specified by the applicable specification sheet to the cut-off voltage specified. The battery shall meet the requirements of 3.6.1 and 3.6.3.

4.7.11 Cycle life. The battery shall be tested as follows:

a. Cycle 1. Charge per 4.6.1.2.1. Rest for one hour at open circuit. Then discharge at the normal rate specified by the applicable specification sheet to the cut-off voltage specified.

b. Cycle 2. Charge per 4.6.1.2.1. Rest for one hour at open circuit. Then discharge at the high rate specified by the applicable specification sheet for 2.0 minutes.

c. Cycle 3-98. Charge per 4.6.1.2.2 and rest for 1 hour at open circuit and discharge at the high rate specified by the applicable specification sheet for 2.0 minutes.

d. Cycle 99. Charge per 4.6.1.2.2 and rest for one hour at open circuit and discharge at the high rate specified by the applicable specification sheet to the cut-off voltage specified.

e. Cycle 100. Repeat step a.

f. Cycle 101. Repeat step b.

g. Cycle 102 -198. Repeat step c.

h. Cycle 199. Repeat step d.

i. Cycle 200. Repeat step a.

NOTE: Water may be added on the 100th cycle. The amount of water added shall be recorded. Reconditioning per 4.6.1.3 is not allowed during the cycle life test.

4.7.11.1 Failure. If any of the following conditions occur during the cycle test, the battery shall be judged to have failed:

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a. The voltage during discharge is below the cut-off voltage specified in the applicable specification sheet.

b. The capacity of either the 100th or 200th cycle is less than the capacity specified on the applicable specification sheet.

c. The battery or cell when inspected per 4.7.5 fail to meet the requirements of 3.4, 3.5, and 3.6.3.

4.7.12 Mechanical shock. The battery shall be charged per 4.6.1.2.1 and then subjected to a Mechanical Shock Test per Procedure I, Method 516.3 of MIL-STD-810 using Figure 516.3-1. The test sample shall be mounted and shocked in the normal upright position. During test the open circuit voltage shall be monitored. The battery shall conform to the requirements of 3.6.2.

4.7.13 Vibration. The battery shall be charged per 4.6.1.2.1 and then subjected to a Vibration Test per Method 514.3, Procedure I and Condition I-3.2.6 of MIL-STD-810. Use the test time schedule and curve for equipment mounting configuration "without vibration isolators". The battery shall be discharged at the rate specified by the applicable specification sheet to the cut-off voltage specified and shall meet the requirements of 3.6.1 and 3.6.2. The open circuit voltage shall be monitored during test.

4.7.14 Temperature shock. The battery shall be charged per 4.6.1.2.1 and subjected to the Thermal Shock Test per Procedure I, Method 503.2 of MIL-STD-810. Upon completion the battery shall be stabilized at normal conditions per 4.6.1.1 and discharged at the rate specified by the applicable specification sheet to the specified cut-off voltage and shall meet the requirements of 3.6.1 and 3.6.2. The open circuit voltage shall be monitored during test. The components (e.g. thermistors, thermoswitches and etc.) of the harness in the battery shall be monitored during the Thermal Shock Test. The components shall meet the performance requirements specified in 3.4.3.

4.7.15 Altitude. The battery shall be tested as follows:

a. Charge per 4.6.1.2.1

b. Place battery with cover in place (on open circuit) into environmental chamber and within 15 minutes lower chamber pressure to simulate pressure at 6000 feet above sea level.

c. Discharge at the rate specified by the applicable specification sheet to the specified cut-off voltage. The battery shall meet the requirements of 3.6.1.

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d. Charge per 4.6.1.2.2 under the conditions of step b.

e. Discharge per step c.

f. After discharge return the chamber to normal conditions specified in 4.6.1.1 within 5 minutes.

g. The battery shall be inspected for defects per 4.7.5 and shall meet the requirements of 3.6.2.

4.7.16 Humidity and charge retention. The battery shall be charged per 4.6.1.2.1 and then subjected to the humidity test per Procedure II, Method 507.2 for 15 cycles of hot-humid conditions of MIL-STD-810. The battery case air vents shall be positioned where the axis of the vents are parallel to the chamber air flow. Upon completion, accumulated water may be removed from the battery. The battery shall be discharged at the rate specified by the applicable specification sheet to the specified cut-off voltage and shall meet the requirements of 3.6.1 and 3.6.2.

4.7.17 High rate and low temperature discharge. The battery shall be charged per 4.6.1.2.1 and stored at -22±2°F (-30± 1.1°C) for 16 hours. The battery shall be discharged at the rate specified by the applicable specification sheet to the specified cut-off voltage and shall meet the requirements of 3.6.1 and 3.6.2.

4.7.18 High rate discharge and operating position. The battery shall be charged per 4.6.1.2.1 and discharged at the rate specified by the applicable specification sheet to the specified cut-off voltage. The battery shall be in an inverted position for the first 2.5 minutes of discharge and shall meet the requirements of 3.6.1 and 3.6.4.

4.7.19 Internal pressure test. Cells shall be subjected to the Internal Pressure Test (para 4.6.22) of MIL-B-81757. The cells shall meet the requirements of 3.4.2.4.

4.7.20 Receptacle mounting test. The battery terminal shall be subjected to the Receptacle Mounting Test (para 4.6.23) of MIL-B-81757. The battery terminal shall comply with 3.4.1.4.

4.7.21 Alkali resistance of harness. The harness assembly, with the exception of the MS3474-L12-10PN connector, shall be immersed in a covered vessel containing 1.300±0.025 specific gravity KOH electrolyte. The connector shall be on the outside of the covered vessel containing the solution and protected from contact with either the solution or vapors from the solution. The vessel shall be stored in a chamber for 7 days at a constant temperature of 150±5°F (64.5±3.3°C). At the end of storage, the harness shall be rinsed with water and dried. The harness assembly shall be inspected for cracks or blisters, electrical continuity and

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isolation (where applicable) and shall meet the requirements of 3.3.1, 3.3.2, 3.6.6 and 3.6.2.1.

4.7.22 Insulation resistance of harness. The insulation resistance of the harness assembly shall be measured between the metal area of the sensor receptacle (exclude applicable pins) and any current carrying part in accordance with MIL-STD-202, Method 302, Test Condition B for one minute and shall meet the requirement of 3.4.3.5.

4.7.23 Harness temperature shock. The harness assembly shall be subject to the Thermal Shock Test per Procedure I, Method 503.2 of MIL-STD-810. Each thermistor and thermoswitch shall be monitored. The harness assembly shall meet the requirements of 3.6.2.1.

4.8 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First Article Inspection of Packaging.

b. Quality Conformance Inspection of Packaging.

4.8.1 First Article Inspection of Packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.8.2 Quality Conformance Inspection of Packaging.

4.8.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.8.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.8.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table IX, "PACKING INSPECTION PROVISIONS." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality Level of 4.0 percent defective.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall

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be in accordance with Special Packing Instruction AG00000071.

6. NOTES

6.1 Intended use. Battery, BB-716/A rechargeable vented, nickel-cadmium, is used in aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Level A or B preservation and packing (see section 5).

c. When first article inspection rough handling tests are not required.

d. If first article inspection (see 3.2), first article test plans and test report(s) are required (see 4.4.1).

e. When certification of parts, materials, or components is not required (see 3.3).

f. Actions are required relative to group C inspection failure (see 4.5.6).

g. When first article packaging inspection report require Acquisition Activity approval prior to production unit packing.

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Definitions.

6.4.1 Fully charged batteries. Batteries shall be considered fully charged when charged in accordance with 4.6.1.2.1.

6.4.2 Discharged batteries. Batteries shall be considered discharged when a specified discharge rate was conducted until a specified final voltage was reached. For shipping purposes only, the battery shall be discharged to zero volts per cell.

6.4.3 End voltage. The end voltage of the battery is the specified end voltage terminating a specified discharge rate.

6.4.4 Cycle. A cycle is a combination of a charge and discharge.

6.4.5 Normal upright position. Perpendicular to the plane of

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the horizon.

6.4.6 Fully formed. Batteries and cells shall be considered fully formed if they have been charged to the point where all available cadmium oxide has been converted to cadmium metal and both hydrogen and oxygen gases were evolved at close to stoichiometric ratios.

6.5 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government may be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product (see 4.2).

6.6 Test report data. The following requirement shall apply to First Article and Quality Conformance Test Reports:

Whenever there is a service or ampere-hour requirement, the following data shall be included:

1. Initial Open Circuit Voltage (IOCV).

2. Initial Closed Circuit Voltage (ICCV).

3. Initial Current Drain.

4. Closed Circuit Voltage and Current shall be recorded at regular intervals. Interval shall be service requirement or shorter.

5. The time elapsed between ICCV and end voltage (EV).

6. Record how a battery is discharged. Particularly whenever near zero volts.

7. The time elapsed between start of charge and 1.55 volts/cell is reached.

Date and time of day that any portion of test is initiated shall be recorded in test report including initiation of soak at test temperature. When a requirement limit is specified, a final quantitative figure shall be recorded.

6.7 International standardization. Certain provisions of this specification are the subject of international standardization agreement (ASCC AIR STANDARD 12/15 and STANAG 3514). When amendment, revision or cancellation of this specification is proposed which will effect or violate the international agreement

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concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

6.8 First Article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of four units. The contracting officer should include specific instructions in the acquisition documents regarding arrangements for examinations and approval of the first article test report.

6.9 Subject term (key word) listing.
Battery
Cells
Harnesses
Nickel-Cadmium

Preparing Activity:
Army-ER

(Project 6140-A601)

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Table I. Environmental Tests

Tests	Test Paragraphs
Mechanical Shock	4.7.12
Vibration	4.7.13
Temperature Shock	4.7.14
Altitude	4.7.15
Humidity and Charge Retention	4.7.16

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Table II. Certification of material/component specification requirements

Material or components	Applicable specifications
Nickel plating	QQ-N-290
Machine screws, studs and nuts	FF-N-836 or FF-S-92
Connectors, Intercell	QQ-C-502 or QQ-C-576
Web Nylon	MIL-W-4088 and 3.4.10
Thermistor	MIL-T-23648 and 3.4.3.1
Thermoswitches	MIL-S-24236 and 3.4.3.3
Wiring	MIL-W-16878 and 3.6.7
Potting	MIL-P-47104, MIL-I-16923 and 3.6.6
Diode	MIL-S-19500 and 3.4.11
Receptacle	MS3509 and MS3474-L12-10PN
Molded plastic parts	3.3.2, MIL-M-14, type MFE
Elastomers	3.3.3
Polyamide material	3.3.2, ASTM D4066-82, Classes PA312L02B00000, PA612, PA111, and PA811.

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Table III. First article inspection of components and materials

Test	Part	Requirement Paragraph	Test Paragraph
Corrosion resistance	Metal parts 2/	3.3.1	4.7.1
Alkali absorption of plastic materials	Plastic parts	3.3.2	4.7.2
Alkali resistance of elastomer materials	Elastomers	3.3.3	4.7.3
Alkali resistance of harness 1/	Harness	3.3.1, 3.3.2, 3.6.6, 3.6.2.1	4.7.21
Insulation resistance of harness 1/	Harness	3.4.3.5	4.7.22
Temperature shock of harness 1/	Harness	3.6.2.1	4.7.23
Venting test	Vent plug	3.4.2.5	4.7.4.1
Torque test	Vent plug	3.4.2.5	4.7.4.2

NOTES: 1/ Test conducted except for cell only procurements.

2/ Container, cover and latches, external cell terminal hardware; intercell connectors and hardware, etc.

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Table IV. First article inspection of batteries

Test	Sample or Cell Group Number 1/	Requirement Paragraph	Test Paragraph
Visual and Mechanical	1-4	3.4, 3.5, 3.6.3	4.7.5
Dimension and Weight	1-4	3.4	4.7.7
Insulation Resistance 2/	1-4	3.6.5	4.7.6
Receptacle Mounting 2/	1-4	3.4.1.4	4.7.20
Container Body to Container Cover 2/	1-4	3.4.1.1	4.7.8
Internal Pressure 3/	1-4	3.4, 3.4.2.4	4.7.19
Capacity (Initial)	1-4	3.6.1, 3.6.3, 3.4.9.1, 3.4.9.2	4.7.10.1
Potassium Carbonate 3/	1-4	3.4.2.6	4.7.9
High Rate Discharge, Low Temperature	2,4	3.6.1, 3.6.2	4.7.17
High Rate Discharge Operating Position	1,3	3.6.1, 3.6.4	4.7.18
Cycle Life	1,2	3.6.1, 3.6.3	4.7.11
Mechanical Shock	3,4	3.6.2	4.7.12
Vibration	3,4	3.6.1, 3.6.2	4.7.13
Temperature Shock	3,4	3.6.1, 3.6.2	4.7.14
Altitude	3,4	3.6.1, 3.6.2	4.7.15
Capacity	3-4	3.6.1, 3.6.3	4.7.10
Humidity and Charge and Retention	3,4	3.6.1, 3.6.2	4.7.16
Capacity	3-4	3.6.1, 3.6.3	4.7.10

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NOTES: 1/ Cell group (see 4.4.2.1)
2/ Not applicable to cell procurements
3/ For battery or battery plus component procurements test four cells and one low capacity cell per battery sample. For separate cell procurement test four cells per cell group.

Table V. Group A Inspection

Test	Requirement Paragraph	Test Paragraph
Visual and Mechanical	3.4, 3.5, 3.6.3	4.7.5
Insulation Resistance 1/	3.6.5	4.7.6

NOTES: 1/ Not applicable to cell procurements.

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Table VI. Identification Defects

Number	Description	Method of inspection
1	Electrical contact surfaces obstructed by insulation compounds.	Visual
2	Pitting or blow holes on the external cells container.	Visual
3	Electrolyte leakage and quantity of electrolyte.	Visual
4	Location and polarity of terminals not as specified.	Visual
5	Terminal, identification, caution and warning markings not as specified.	Visual
6	Terminal seal missing or defective.	Visual
7	Corrosion.	Visual
8	Particles of foreign material.	Visual
9	Insulators or insulation missing or damaged.	Visual
10	Welds containing blow holes, cracks, or slag inclusions.	Visual
11	Burrs on battery container, cover or interconnecting hardware.	Visual
12	Improper color on outside of container and cover.	Visual
13	Thermal switches open at room temperature.	Meter
14	Incorrect thermistor values.	Meter
15	Date codes missing or incorrect or more than one date code on a battery.	Visual
16	Carrying strap on cover missing, twisted or torn.	Visual
17	MS3474L12-10PN mounted incorrectly (key not oriented at 12 o'clock position).	Visual

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Table VII. Group B Inspection

Test	Requirement Paragraph	Test Paragraph	Inspection Level	AQL
Dimension and Weight	3.4	4.7.7	S-3	2.5%
Container Body to Container Cover 1/	3.4.1.1	4.7.8	S-3	2.5%
Internal Pressure 2/	3.4, 3.4.2.4	4.7.19	S-3	2.5%
Vent Plug Test 2/	3.4.2.5	4.7.4	S-3	2.5%
Capacity (Initial)	3.6.1, 3.6.3 3.4.9	4.7.10.1	S-3	2.5%
Subgroup II				
High Rate Discharge, Low Temperature	3.6.1, 3.6.2	4.6.17	S-3	1.5%
High Rate Discharge, Operating Position	3.6.1, 3.6.4	4.7.18	S-3	1.5%

NOTES: 1/ Not applicable to cell procurements.
2/ For battery or battery plus componentprocurements, test four cells and low capacity cell per batterysample. For separate cell procurement test four cells per cell group (see 4.5.2.1).

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Table VIII. Group C Inspection

Test	Requirement Paragraph	Test Paragraph	Subgroup I	Subgroup II
Potassium Carbonate 1/	3.4.2.6	4.7.9	X	X
Capacity	3.6.1, 3.6.3	4.7.10	X	X
Humidity and Charge Retention	3.6.1, 3.6.2	4.7.16	X	
Cycle Life, 200 cycles	3.6.1, 3.6.3 3.6.8	4.7.11		X
Temperature Shock 2/	3.6.1, 3.6.2 3.4.3, 3.6.2.1	4.7.14	X	
Vibration 3/	3.6.1, 3.6.2	4.7.13	X	
Alkali Resistance of Harness 3/	3.3.1, 3.3.2 3.6.6, 3.6.2.1	4.7.21		X

NOTES: 1/ For battery or battery plus component procurements, test 4 cells and 1 low capacity cell per battery sample. For separate cell procurement test 4 cells per cell group (see 4.5.2.1).
2/ For separate harness assembly procurements (see 4.7.23) for test method.
3/ Not applicable for cell procurements.

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Table IX. PACKING INSPECTION PROVISIONS

Number	Characteristic	Method of Inspection
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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